

Institute of Actuaries of India

Subject SA5 – Finance

May 2014 Examinations

INDICATIVE SOLUTIONS

Solution 1 : (i)

W.P.I	C.P.I
Wholesale price index measures inflation at each stage of production	Consumer price index measures inflation only at final stage of production
Wholesale price index is the middle point of the sum of all the goods bought by the traders	Consumer price index is the middle point of the sum of all the goods bought by consumers
Wholesale Price Index (WPI), is based on the price prevailing in the wholesale markets or the price at which bulk transactions are made	The Consumer Price Index (CPI), is based on the final prices of goods at the retail level.
There are only few countries that uses WPI to calculate inflation rates	Many nations have already shifted to using CPI.

- As may be seen from above, the CPI reflects the price the end consumer is paying and hence the true cost of living.
- The gap between prices at wholesale level and retail level could be attributed to supply chain bottlenecks which impede the process of smooth distribution of goods. Post-production hurdles at several levels including storage, packaging, marketing and infrastructure is a major fuel to retail prices
- The high weightage of food prices in calculation of CPI. If the food prices go up, CPI may be much higher than the WPI.

[3]**(ii)**

- The primary instrument which RBI uses to control inflation is through its monetary policy wherein it sets the benchmark interest rates at which the banks can borrow from RBI or lend to RBI
- Higher rates make borrowing more expensive and saving more attractive. This should lead to lower growth in consumer spending and investment

- A higher interest rate should also lead to higher exchange rate, which helps to reduce inflationary pressure by
 - making imports cheaper.
 - Reducing demand for exports and
 - Increasing incentive for exporters to cut costs.
- RBI can advise the government on fiscal measures to be taken but does not have direct control over it.
- RBI can look at increasing the risk capital weightages for sectors where there could be more lending is happening. For e.g.: if too much money is being lent to real estate sector, which may be adding to inflation or artificial jack up of housing prices, RBI may increase the risk capital requirement for loans given to this sector.
- RBI may look at increasing the CRR & SLR in order to control money supply as well which can make lending costlier to control speculative investments
- RBI will advise government to help ease the supply side constraints as these policies may enable the economy to become more competitive and help to moderate inflationary pressures. For example, more flexible labour markets may help reduce inflationary pressure. However, supply side policies can take a long time, and cannot deal with inflation caused by rising demand.
- RBI will require government support to control inflation as solely looking at monetary policy measures may lead to decline in investments, which may have long term adverse economic consequences such as lower growth, un-competitiveness of the prices of goods leading to lower exports etc. **[6]**

(iii)

- RBI uses monetary policy by calibrating the bench mark interest rates to combat inflation. Therefore if the inflation is high, the bench mark interest rates may be increased to make money dearer
- The interest rates could be higher if the demand for investment goes up due to supply side gaps. As the money is limited, the interest rates may go up if the demand for investment rises.
- The interest rates could also go up, if the country is facing forex crunch as the RBI may increase the interest rates in order to attract foreign exchange.
- In India, even though currently, the interest rates are high mainly due to inflation, which itself is a direct results of inadequate supply of goods, high food prices, high oil prices (as we depend largely on crude imports) etc.,
- In the past in India, high interest rates were prevalent due to inadequate forex.

You are the Appointed Actuary of a medium sized life insurance company in India. In that capacity you are also a member of the Investment Committee of the company.

In the forthcoming Investment Committee meeting, the Chairperson of the Investment Committee wants you to present your views on the recent trends in the Indian economy and how these trends would impact the short to medium term outlook for the Indian life insurance industry. [4]

(iv)

Key themes related to the recent trends in the Indian economy:

Macroeconomic conditions deteriorated during the last three to four years, posing several challenges .

- Growth slowed down to a 10 year low of 5.0% during 2012-13. This slow down has been pervasive across sectors including services. Over the past two years, although part of the slowdown has been driven by cyclical factors, structural constraints have played a major role in the slowdown. Mining activity was impacted adversely by governance factors. The slack in manufacturing activity was largely due to poor investment on the back of structural issues facing the infrastructure sector.
- In addition, global factors played an important role in the current growth slowdown. Global growth decelerated to 3.1 per cent in 2012, the lowest since the 2009 contraction that followed the global financial crisis. Consequently, external demand fell and revival was difficult. Two years of industrial slowdown and dampened demand has slowed down services sector activity as well.
- Though growth began decelerating in the recent years, inflation stayed near double digits. This led to the Reserve Bank keeping the monetary policy in tightening mode until December, 2011. Thereafter with signs that inflation was moderating in line with projections and with demand-side pressures starting to ebb, the Reserve Bank started easing the monetary policy through successive reductions in CRR, cuts in the policy rate, and imparting liquidity through OMOs [Open Market Operations]. The easing of monetary policy to provide a more pro-active growth-supportive stance has been constrained by widening fiscal and current account deficits and inflation persistence
- One of the key macro economic challenges facing the Indian economy has been the widening gross fiscal deficit [GFD] in the recent years. There has been a fiscal correction during 2012-13 to contain the GFD to 4.9 per cent of GDP from 5.7 per cent in 2011-12. In the process, the actual fiscal deficit in 2012-13 turned out to be lower than that envisaged in the Union Budget.

- It is worth noting that the fiscal correction during 2012-13 in the face of a shortfall in tax and non-tax revenues was largely brought about by significantly scaling down capital expenditure. In fact, the capital outlay to GDP ratio was lower in 2012-13 than in 2011-12. While the reduction in planned capital expenditure may have been dictated by the immediate need to reduce the headline deficit, it does raise concerns about the quality of fiscal consolidation. Over the medium term, efforts should be to contain revenue expenditure, raise tax revenue buoyancy and contain subsidies to enable a more durable fiscal consolidation.
- External sector vulnerabilities came to the fore in 2012-13, as the CAD [Current Account Deficit] widened to a peak of 4.8 per cent of GDP. The widening of the CAD was largely the result of high oil and gold imports and moderation in export growth. In order to contain gold imports, import duties on gold were increased in a phased manner from 2 percent to 8 percent. With a view to restraining fiscal subsidies and to allow price responses to work to curtail demand for oil and help bring about a CAD correction, the government hiked the price of diesel and capped the supply of subsidised LPG cylinders in September 2012. Later in January 2013, it partially deregulated diesel prices by allowing a monthly reset. These measures are expected to help contain oil demand to some degree. However, oil and gas prices are yet to be fully market-determined and in the absence of free pricing the reduction in demand may not be enough to shrink the trade deficit to the desired extent.
- With slowing growth over the past two years, external sector sustainability has also emerged as a key concern on the horizon. India's external debt has steadily increased over the last two years. Also the short-term external debt constituted 44 per cent of total external debt and 59 per cent of foreign exchange reserves at end-March 2013. Thus, maintaining external sector sustainability poses an important challenge going forward.
- Indian economy is currently going through a difficult period. What is important at this stage is to preserve India's growth potential by arresting the downtrend and maintaining stable macroeconomic conditions. For this, the focus needs to be on implementation of measures aimed at removing structural constraints so that production and investment activity could gather momentum. Current slowdown has been accentuated by structural factors that have come in the way of smooth adjustment through pure demand management policies.
- With consumer price inflation, fiscal deficit and current account deficit being at higher than their optimum levels, the need to preserve macroeconomic stability has emerged as a binding constraint. As such the momentum of recovery could come from reengineering focus on unclogging the stalled investment projects, giving an impetus to investments in key infrastructure sectors, supporting productivity enhancements by technology

enhancements, bringing in more managerial efficiencies and supporting research and development.

- Inherently, the Indian economy has several strengths including its natural endowment and demographic dividends. Simple institutional reforms such as better regulation of natural resources, improved harnessing of water resources, investing more in skill formation, digitalizing land records, land consolidation, better integration of regional agricultural markets, freer labour markets and more competitive domestic markets can go a long way in improving India's potential as well as actual growth. As such, efforts in the direction of macroeconomic stability and structural reforms can pave the way for the recovery.

Implications for the short to medium term outlook for the Indian life Insurance industry.

[a] Fall in disposable income on account of high price inflation is likely to dampen the demand for savings oriented life insurance products on the grounds of premium being unaffordable. There is likely to be an increase in the demand for pure protection products.

[b] The widening current account deficit and the external sector challenges are likely to inhibit stable foreign capital flows. This can imply an inherent reluctance on the part of the foreign insurers to hike their equity investment / invest in new insurance ventures even if the FDI limit is hiked to 49%.

[c] The revival of demand for life insurance products hinges crucially on the government's efforts to surmount the twin challenges of correcting the fiscal and current account deficits.

[d] The high interest rates on short term deposits can render traditional life insurance products relatively less attractive.

[e] Given the volatile conditions in the stock market, the unit-linked products can be perceived to be riskier.

[f] Higher than expected expense inflation are likely to shrink the expected profit margins on the life insurance products.

In the Investment Committee meeting, after hearing your presentation, one of the members makes a comment that "the high interest rates prevailing in the Indian economy are good for life insurance companies".

[12]

(v) a. Investment of Assets:

Life Insurance liabilities are long term in nature. The assets are invested to match the liabilities by term & nature in order to meet the expected outgoes.

With current regime of higher interest rates, the assets can be invested in high yielding securities which would be provide higher coupon investment income for a longer term.

Typically the assets are invested to hold to maturity. Therefore even though, the market value of assets comes down with higher interest rates due to unrealized losses, it would not impact the cash flows as long as assets are held to maturity.

b. Product Pricing:

If the high interest rate regime is expected to continue, the pricing of products can be done based on the expected yield on the assets. If the yield is higher, the premium rates would be cheaper. This would be especially useful while pricing single premium and very limited pay products as assets can be locked in quickly and the risk of reinvestment is limited to the coupons rather than the renewal premiums.

In case of non-participating products, the investment risk is completely borne by the Company/Shareholders. Therefore while setting the interest rate for pricing of these products, the long term interest rate view should be taken as the future premiums received would need to be invested at the then prevailing interest rates. If the interest rates fall, the inherent guarantees of non-participating products may become onerous.

Therefore if it is decided to price non-participating products at the current prevailing interest rates, it would be more prudent to go for limited pay products. In case of regular premium products, the Company should be in position to re-price the product immediately, if the interest rate trends start reversing.

In case of participating products, high interest rates may not necessary lead to lower premiums, since the pricing can be done with long term interest rates and if the interest rates go up, the bonuses can be increased in line with the actual investment earnings. Similarly if the interest rates go down, the bonuses would be reduced. On the other hand, if we choose to price these products at the prevailing rates, then the premiums may be cheap but the ability to increase the bonuses in future may reduce. Hence the companies take a balanced view while setting the interest rate after taking into consideration the current economic situation and the likely trend in interest rates.

c. Valuation of Liabilities:**Non-Participating Products:**

If the risk free rates are high, the value of non-participating liabilities would come down thus easing the capital requirements. The actuary while setting the valuation interest rates takes into account the guidance notes issued by the Institute as well as the relevant regulations issued by the IRDA. Typically the valuation interest rate is set taking the bench mark 10 year rates as the basis for deciding the risk free rate as well as the margin for adverse deviation.

If the trend of high interest rates continues, the non-participating liabilities would come down, if the actuary decides to increase the discount rate based on the prevailing market rates.

As a result, Solvency is expected to improve as the liabilities would be lower with higher interest rates, whilst the assets are valued at book value, without taking into consideration the market value.

As stated above, the opposite would happen if the interest rates decline.

Participating Products:

High interest rates would not impact the liabilities much as with higher discount rates, the corresponding bonuses also have to be increased thus the positive of higher discount rate is compensated by higher liabilities due to higher bonuses. Therefore the interest rate on participating products would not have much effect. Conversely, when the interest rates fall, the future bonuses also would be reduced thus having minimal impact on liabilities.

[10]

(vi)

In the Indian context the primary sources of capital are as follows :

Equity from shareholders & the Profits made in the business

Currently the undistributed surplus pertaining to participating policyholders also can be a source of capital for demonstrating solvency. However this surplus cannot be made use of for distributing dividends.

Reinsurance – Primarily surplus reinsurance .Currently upto a maximum of 15% of reserves and 50% of Sum at Risk can be taken credit towards reinsurance while computing the solvency requirement in India.

In the global context, life insurance companies can raise both equity and debt capital to meet their capital requirements

The financing options [other than raising equity capital] include:

- Utilizing free assets not allocated to policyholders (the inherited estate)
- Issuing subordinated debt or hybrid capital
- Raising capital secured on the future profits expected to arise from in force business. This can be done through reinsurance financing (financial reinsurance); bank contingent debt or securitization
- Other forms of reinsurance financing (e.g.: original terms quota share)

In the case of a mutual company, demutualization can be way of raising equity capital from a third party or the market.

In the case of a proprietary life company which is a subsidiary of a holding company (e.g.: a bank), the holding company can raise capital and pass the capital down to the life insurance subsidiary, leaving the subsidiary free of any obligation to repay the capital

The capital of a life insurance company is tiered based on its loss absorbency and performance. Typically it is three tiered:

- Tier 1 capital is of the highest quality and is the most loss absorbent and permanent form of capital. Examples will included paid up ordinary share capital and disclosed reserves.
- Tier 2 capital ranks next and will include sources of capital like hidden reserves unrealized gains on investment securities, medium to long term subordinated debt, and financial reinsurance.
- Tier 3 is of the lowest quality from the stand points of loss absorbency and permanence. Example of Tier 3 capital can be short term subordinated debt.

[5]

(vii)

- Solvency II is based on three pillars. Pillar 1 involves the calculation of SCR (Solvency Capital Requirement) and MCR (Minimum Capital Requirement). Pillar 2 focuses on governance issues and risk management procedure. The ORSA [Own Risk and Solvency Assessment] report provided to the regulator will form a key part of this pillar. Pillar 3 involves disclosure and reporting requirements including more disclosures to the market.
- On the other hand, Solvency I is a single pillar involving only the calculation of SCR.
- As compared to Solvency I, Solvency II is a more risk-based regime where insurers are required to be more sophisticated in their modeling of risk and return. Therefore Solvency II's capital requirement tends to be a better estimate of the capital required to cover the risks.
- Solvency I is based on "one size fits all" approach. On the other hand, Solvency II allows different insurers to adopt different methods of modeling risk, making the resulting capital more entity specific.

- Solvency II covers more risks than Solvency I. Solvency II covers market, credit, liquidity and operational risks in addition to insurance risks. Also Solvency II has tighter asset recognition rules.
- Solvency II requires more elaborate and open disclosures to the market rather than just to the regulator. Insurance companies have to draw up a “ORSA” which discloses a great deal about their risk assessment and capital framework. The ORSA report needs to be presented to the regulator. In addition, the insurance companies are required to make public, their solvency and financial conditions reports.

[4]

(viii) Valuation of Liabilities:

Non-Participating Products:

If the risk free rates are low, the value of non-participating liabilities would increase thus increasing the capital requirements. Typically the valuation interest rate is set taking the current bench mark 10 year rates plus the margin for adverse deviation which means that the valuation rate of interest will be lower than the prevailing rate.

As a result, Solvency is expected to worsen as the liabilities would be higher with lower interest rates, whilst the assets are valued at book value, without taking into consideration the market value. Had the assets [which would predominantly be GoI bonds and AAA rated corporate bonds] been valued at market prices, the asset values would also be higher thus counteracting the negative impact on solvency ratio.

Participating Products:

Lower interest rates would not impact the par liabilities as much as it would impact the non-par liabilities. This is because the corresponding future reversionary bonuses will also be decreased thereby partly offsetting the increase in liabilities caused by the lower valuation rate of interest.

[7]

Assume that the solvency capital requirement [SCR] for the life insurance company has to be estimated using the standard formula approach prescribed under Solvency II framework.

a) The Solvency Capital Requirement (SCR) is a Value at Risk measure based on a 99.5% confidence interval of the variation over one year of the amount of “basic owned funds” (BoF). Broadly BoF is equal to assets minus technical provisions.

There is a prescribed list of risk groups which the SCR has to cover which includes:

Life underwriting risk

Non-life underwriting risk

Health underwriting risk

Market risk

Counterparty default risk

Operational risk

Intangible asset risk

Market risk is further classified into sub modules such as

Equity risk

Property risk

Interest rate risk

Spread risk

Currency risk

Concentration risk

Illiquidity risk

For each of these sub modules of market risk, the SCR can be assessed using either a standard formula approach or an internal model approach.

Under the standard formula approach the SCR for each of the sub modules of market risk is calculated using standard prescribed stress tests or factors.

The SCR for market risk is calculated by aggregating the SCR for all sub modules using prescribed correlation matrices.

For example the capital requirements for listed equities may be calculated as the loss in the basic owned funds (BoF) that would result from an instantaneous decrease of, say, 40% in the value of such equity investments. Likewise the capital requirement for property risk will be calculated as a loss in the BoF that would result from an instantaneous decrease of ,say, 25% in the value of real estate.

Once the SCR's for all sub modules of market risk have been calculated the SCR for market risk will be calculated using the following formula:

$$SCR[MarketRisk] \\ = [\sum Corr(i, j) \times SCR(i) \times SCR(j)]^{0.5}$$

Where

The sum covers all possible combinations (i,j) of the sub modules of the market risk module

Corr (i,j) denotes the correlation coefficient between sub modules I and j

SCR (i) and SCR (j) denotes the capital requirements for sub modules I and j respectively.

An alternative to the standard formula approach is to use the internal model approach for calculating the SCR. This is particularly appropriate if the market risk profile of the insurer differs materially from that covered by the standard formula and/or if the company already uses such a model for risk management or for other decision- making purposes [e.g.” setting investment strategy].

The capital requirement resulting from the use of an internal model will typically differ from the outcome of the standard formula calculation. It can be either higher or lower depending on how the company’s risk profile compares with the assumptions underlying the standard formula.

However the internal model must still generate an SCR based on stated requirements including coverage of all relevant sub risk modules and providing at least the equivalent protection to a 99.5% confidence interval over one year.

The internal model must also be approved by the regulator.

[6]

b) The capital requirement for interest rate risk will be equal to the larger of the following:

The capital requirement for the risk of an increase in the term structure of interest rates.

The capital requirement for the risk of decrease in the term structure of interest rates

The capital requirement for the risk of an increase in the term structure of interest rates will be equal to the loss in BoF that would result from an instantaneous increase in risk-free interest rates at different maturities in accordance with a specified table. For example, a 70% increase may be specified for maturity of 1 year; 65% increase may be specified for a maturity of 2 years and so on.

The capital requirement for the risk of a decrease in the term structure of interest rates will be equal to the loss in BoF that would result from an instantaneous decrease in risk free interest rates at different maturities again in accordance with a specified table. For example, a 75%

decrease may be specified for maturity of 1 year; 70% decrease may be specified for a maturity of 2 years and so on. [3]

[Total Marks-60]

Solution 2 :

i) Some of the plausible reasons for merger and acquisition are as under:

1. to benefit from economies of “vertical integration” – this means merging with a supplier or customer to facilitate coordination and reduce expenses
2. to benefit from complementary resources
3. to optimize use of tax breaks
4. To eliminate inefficiency
5. to benefit from opportunities that might only be available to larger companies
6. protection against threat of takeover – by increasing the size
7. utilization of surplus funds
8. enhancement of earnings per share
9. lower financing cost

[6]

ii) **Assessment of net cash flow:** Water Melon needs to assess benefits accruing to its equity shareholders, based on following information:

1. estimation of future profits, plus depreciation, available to equity shareholders, net of tax
2. The current value of all surplus assets not required to earn profits. If the target company has excess assets not required to back the company’s operations then these can be used for other purposes, or returned to shareholders.
3. the replacement cost of non-surplus assets, and the estimation of when replacement will be required, for example, replacement of machinery
4. the amount, and timing, of any loan or debt redemption
5. All contractual obligations with respect to short and long term debt also needs to be scrutinized.

Approach for Valuing the Net Cash flows associated with Acquisition:

One of the commonly used approaches for valuing an acquisition is the Free Cash Flow Approach. The steps involved in this approach are as follows:

1. Define the present value of the free cash flows from the acquisition: The free cash flow [FCF] arising from an acquisition is the cash flow generated by it after providing for the additional investment required for its operations. Typically FCF is expressed as: Net operating profit adjusted for taxes less Net Investment. The Net operating profit adjusted for taxes will be equal to $EBIT \times (1 - \text{Tax rate})$. The Net Investment is equal to Gross Investment in Fixed Assets and Current Assets less depreciation.

The FCFs are projected over an explicit forecast period [N years] after which the growth rate in free cash flows is expected to be constant. The present value of these free cash flows over the N years is determined using a discount rate which reflects the opportunity cost of capital to the equity shareholders assuming that the acquisition is completely financed through equity.

2. The second step is to estimate the present value of the horizon value beyond the explicit forecast period. Assuming that the constant growth rate in free cash flows to be “g” pa beyond the explicit forecast period of N years, the horizon value can be estimated using the constant dividend growth model as follows :

$$\text{Horizon Value [HV]} = \text{FCF (N+1)} / (r - g)$$

The Present Value of the Horizon value will be equal to $HV \times \text{PVF}(r, N)$ where PVF denoted the Present Value Factor

3. The value of the acquisition will be equal to Present Value of Free Cash flows Plus Present Value of Horizon Value.

[10]

iii)

a) Earnings Per Share

Advantages :

- Most commonly and widely accepted parameter
- Data availability is high
- Earnings per share reflects prima facie the earning power

Disadvantages :

- EPS can be subject to differences in accounting policies and manipulation
- Unless adjusted, can be subject to one-off exceptional items
- Cannot be used if earnings are negative
- Fails to take into account the growth rate of earnings of the two companies

- Fails to take into account the differential risks associated with the earnings of the two companies
- Fails to take into account the gains in earnings arising out of the merger

b) Market Price Per Share

Advantages :

- Most commonly and widely accepted parameter
- Data availability is high provided the shares of the acquiring firm and the target firm are actively traded in a competitive market.
- Market prices reflect current earnings, growth prospects and risk characteristics

Disadvantages :

- Not available for unlisted companies
- When trading is meagre , market prices may not be very reliable
- Driven by market sentiments
- Market prices can be manipulated by those who have a vested interest

c) Book Value Per Share :

Advantages :

- Can be useful where assets are a core driver of earnings such as capital-intensive industries
- Most widely used in valuing financial companies, such as banks, which rely on a large asset base to generate profits

Disadvantages :

- Book values for tangible assets are stated at historical cost, which is not a reliable indicator of economic value
- Book value for tangible assets can be significantly impacted by differences in accounting policies

We will recommend using EPS as the base for setting as exchange ratio for the following reasons:

1. The company is unlisted and hence will not have a reliable and widely accepted tradable market value

2. Book value is not suitable as they may be valued long time back and may not represent its true intrinsic or fair value on the day of transaction.
3. Book value is also heavily dependent on accounting treatment and methodology used
4. EPS is more suitable for the FMCG business as it reflects the inherent risk of the business.
5. EPS is widely acceptable as key valuation parameter for the FMCG business.

[8]

iv) Some of the guiding principles are as us under :

1. All holders of the securities of an offeree company of the same class must be afforded equivalent treatment; moreover, if a person acquires control of a company, the other holders of securities must be protected.
2. The holders of the securities of an offeree company must have sufficient time and information to enable them to reach a properly informed decision on the bid; where it advises the holders of securities, the board of the offeree company must give its views on the effects of implementation of the bid on employment, conditions of employment and the locations of the company's places of business.
3. The board of an offeree company must act in the interests of the company as a whole and must not deny the holders of securities the opportunity to decide on the merits of the bid.
4. False markets must not be created in the securities of the offeree company, of the offeror company or of any other company concerned by the bid in such a way that the rise or fall of the prices of the securities becomes artificial and the normal functioning of the markets is distorted.
5. An offeror must announce a bid only after ensuring that he/she can fulfill in full any cash consideration, if such is offered, and after taking all reasonable measures to secure the implementation of any other type of consideration.
6. An offeree company must not be hindered in the conduct of its affairs for longer than is reasonable by a bid for its securities.

[5]

v) Brief summary of some of the most important Rules:

- When a person or group acquires interests in shares carrying 30% or more of the voting rights of a company, they must make a cash offer to all other shareholders at the highest price paid in the 12 months before the offer was announced (30% of the voting rights of a company is treated by the Code as the level at which effective control is obtained).

- When interests in shares carrying 10% or more of the voting rights of a class have been acquired by an offeror (i.e. a bidder) in the offer period and the previous 12 months, the offer must include a cash alternative for all shareholders of that class at the highest price paid by the offeror in that period. Further, if an offeror acquires for cash any interest in shares during the offer period, a cash alternative must be made available at that price at least.
- If the offeror acquires an interest in shares in an offeree company (i.e. a target) at a price higher than the value of the offer, the offer must be increased accordingly.
- The offeree company must appoint a competent independent adviser whose advice on the offer must be made known to all the shareholders, together with the opinion of the board.
- Favorable deals for selected shareholders are banned.
- All shareholders must be given the same information.
- Those issuing takeover circulars must include statements taking responsibility for the contents.
- Profit forecasts and asset valuations must be made to specified standards and must be reported on by professional advisers.
- Misleading, inaccurate or unsubstantiated statements made in documents or to the media must be publicly corrected immediately.
- Actions during the course of an offer by the offeree company which might frustrate the offer are generally prohibited unless shareholders approve these plans.
- Stringent requirements are laid down for the disclosure of dealings in relevant securities during an offer.
- Employees of both the offeror and the offeree company and the trustees of the offeree company's pension scheme must be informed about an offer. In addition, the offeree company's employee representatives and pension scheme trustees have the right to have a separate opinion on the effects of the offer on employment appended to the offeree board's circular or published on a website.

[5]

vi) The shareholders may interpret Melon's action in any of the following terms:

1. Melon's profits increased substantially and the directors are confident that the increase is sustainable. The dividend was therefore increased to reflect the increase in long-run sustainable earnings.

2. Melon's profits have increased and even if the increase is not necessarily sustainable, the company has a publicly-stated target pay-out ratio which it wishes to continue to maintain in order to avoid an adverse market reaction from failing to do so.
3. Melon's wishes to keep its dividends in line those of its competitors who have recently increased their dividend pay-outs substantially. The motive here may again be to avoid possible adverse market sentiment from a failure to do so. Melon's directors decided to use a substantial dividend increase to signal their confidence in the future prospects of the company to investors, in the hope that this might lead to an increase in the company's share price.
4. Melon accumulated a large cash reserve and there are no profitable investment opportunities available in the near future on which to spend it. The directors therefore decided that the most appropriate cause of action was to return part of this money to shareholders via increased dividend payments.
5. Melon decided to increase its dividends as one element in a long-term strategy aimed at rebalancing its capital structure away from equity and towards debt.
6. Melon increased its dividends in response to active demands from its major shareholders to do so.
7. There has been a change in the taxation of dividends relative to retained earnings, which means that it is now more attractive to pay dividends to shareholders.
8. There has have been a relaxation of statutory restrictions that previously constrained the level of cash dividends Melon is able to pay.
